MRC Lecture 5 — Derivation Guide 1

The SantaLucia Nearest-Neighbor Model

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1. Motivation

This guide derives the **nearest-neighbor thermodynamic model** for DNA duplex stability and shows that it is a **dot product** in raw sequence space. The model, finalized by SantaLucia (*PNAS*, 1998) after decades of experimental work, provides the 10 tabulated parameters used by every PCR primer design tool worldwide. The derivation is the prototypical example of **given bilinearity** — the binding energy literally decomposes as a sum of per-position contributions, each of which is a product of agent and target features. No machine learning is required for the form; the physics already gives us the dot product.

By the end of this guide, students will have computed the free energy, dissociation constant, and melting temperature of a specific DNA duplex from scratch, and will see that the calculation is mathematically a dot product — exactly the form required by the convergence equation.

2. The Physical Setup

Two complementary DNA single strands associate to form a double-stranded duplex:

$$5\text{'-}s_1 s_2 \ldots s_L\text{-3'} + 3\text{'-}s_L{}' s_{L-1}{}' \ldots s_1{}'\text{-5'} \rightleftharpoons \text{duplex}$$

where $s_i s_i{}' \in \{\text{A-T, T-A, G-C, C-G}\}$ forms a Watson-Crick base pair at position i. For our purposes, the two strands are perfectly complementary (no mismatches).

The question. What is the free energy change ΔG of duplex formation, and how does it depend on the sequence?

The answer (SantaLucia 1998). The free energy decomposes as a sum of **nearest-neighbor** contributions plus an initiation penalty:

$$\Delta G_{\text{duplex}} = \Delta G_{\text{init}} + \sum_{i=1}^{L-1} \Delta G_{\text{NN}}\left(s_i s_{i+1} / s_i{}' s_{i+1}{}'\right)$$

where ΔG_{NN} is the free energy contribution of the dinucleotide stack at positions $(i, i+1)$. A 6-base-pair duplex has 5 nearest-neighbor pairs ($L-1=5$).

3. The Ten Nearest-Neighbor Parameters

By symmetry (each NN pair can be read in two directions: 5'→3' on one strand, 3'→5' on the other), the 16 possible dinucleotides reduce to 10 unique parameters. At 37°C and 1 M NaCl:

NN pair (5'→3' / 3'→5')	ΔG (kcal/mol)	ΔH (kcal/mol)	ΔS (cal/mol·K)
AA/TT	−1.00	−7.9	−22.2
AT/AT	−0.88	−7.2	−20.4
TA/TA	−0.58	−7.2	−21.3
CA/TG	−1.45	−8.5	−22.7
GT/AC	−1.44	−8.4	−22.4
CT/GA	−1.28	−7.8	−21.0
GA/CT	−1.30	−8.2	−22.2
CG/CG	−2.17	−10.6	−27.2
GC/GC	−2.24	−9.8	−24.4
GG/CC	−1.84	−8.0	−19.9

Plus initiation parameters:

- Initiation with terminal G·C: $\Delta G = +0.98$ kcal/mol
- Initiation with terminal A·T: $\Delta G = +1.03$ kcal/mol
- For a duplex with both ends: sum both initiation terms (~+1.96 kcal/mol typical)

Key observations about the table:

- All ΔG values are negative (duplex formation is favorable at 37°C)
- GC-rich pairs (CG/CG, GC/GC) are most stable (~−2.2 kcal/mol) — both stronger hydrogen bonding (3 H-bonds) and favorable stacking
- AT-rich pairs (TA/TA, AT/AT) are least stable (~−0.6 to −0.9 kcal/mol) — fewer H-bonds, less favorable stacking
- The dynamic range is about 1.6 kcal/mol per NN pair

4. Worked Example: A 6-Base-Pair Duplex

The duplex:
5'-GCATGC-3' 3'-CGTACG-5'
This is a palindrome (reads the same on both strands 5'→3'), which simplifies the NN identification.

Step 1: Identify the nearest-neighbor pairs.

Reading the top strand 5'→3' and tracking each adjacent-pair window:

- Positions 1-2: GC (top) / CG (bottom) → **GC/CG** pair
- Positions 2-3: CA (top) / GT (bottom) → **CA/GT** pair (same as GT/AC by symmetry, tabulated as GT/AC)
- Positions 3-4: AT (top) / TA (bottom) → **AT/AT** pair
- Positions 4-5: TG (top) / AC (bottom) → **TG/AC** pair (same as CA/TG by symmetry)
- Positions 5-6: GC (top) / CG (bottom) → **GC/CG** pair

So we have 5 NN pairs: GC/CG, GT/AC, AT/AT, CA/TG, GC/CG.

Note on symmetry. When identifying NN pairs, remember that $\Delta G_{\text{NN}}(XY / \acute{X}\acute{Y})$ equals ΔG_{NN} read in the other direction. So CA/TG and its reverse GT/AC are the same parameter — but the table lists both forms depending on the convention used. For safety, always write out both strands and look up exactly what's there.

Step 2: Look up each parameter.

- GC/CG = **−2.24** kcal/mol (Wait — the table shows GC/GC = −2.24. Let me verify the convention.)

Convention check: "GC/CG" in the 5'→3' / 3'→5' reading means 5'-GC-3' paired with 3'-CG-5'. Reading 3'-CG-5' backwards (i.e., 5'-GC-3'), this is the GC/GC entry in the table. So GC/CG = GC/GC = **−2.24 kcal/mol**.

Using the table with care:

- Position 1-2: 5'-GC-3' / 3'-CG-5' = GC/CG = −2.24 kcal/mol
- Position 2-3: 5'-CA-3' / 3'-GT-5' = CA/TG = −1.45 kcal/mol (tabulated as CA/TG)
- Position 3-4: 5'-AT-3' / 3'-TA-5' = AT/AT = −0.88 kcal/mol
- Position 4-5: 5'-TG-3' / 3'-AC-5' = TG/AC. By the symmetry $\Delta G(TG / AC) = \Delta G(CA / TG)$, this is also −1.45 kcal/mol
- Position 5-6: 5'-GC-3' / 3'-CG-5' = GC/CG = −2.24 kcal/mol

Step 3: Sum the nearest-neighbor contributions.

$$\Delta G_{\text{NN}} = -2.24 + -1.45 + -0.88 + -1.45 + -2.24 = -8.26 \text{ kcal/mol}$$

Step 4: Add the initiation parameters.

Both ends of the duplex are G·C pairs. Using the G·C initiation parameter (~+0.98 kcal/mol per end):

$$\Delta G_{\text{init}} = +0.98 + +0.98 = +1.96 \text{ kcal/mol}$$

Step 5: Total.

$$\Delta G_{\text{duplex}} = \Delta G_{\text{NN}} + \Delta G_{\text{init}} = -8.26 + 1.96 = -6.30 \text{ kcal/mol}$$

5. From ΔG to K_d

Using the relationship from Lecture 4 ($\Delta G = RT \ln K_d$), at $T = 310$ K where $RT = 0.616$ kcal/mol:

$$K_d = \exp(\Delta G / RT) = \exp(-6.30 / 0.616) = \exp(-10.23) \approx 3.6 \times 10^{-5} \text{ M}$$

So $K_d \approx 36$ μM for this duplex at physiological temperature.

Interpretation. A 6-bp duplex is marginally stable under physiological conditions — bound and unbound populations coexist in roughly similar proportions. Longer duplexes (10–30 bp) are much more stable. The K_d scales approximately exponentially with duplex length at constant GC content, which is why PCR primers are typically 18–25 bp long (giving K_d in the nM to pM range — stable at annealing temperatures).

6. Melting Temperature

The melting temperature T_m is the temperature at which half of the duplexes are dissociated. Using the full ΔH and ΔS decomposition (not just ΔG):

$$T_m = \frac{\Delta H}{\Delta S + R \ln(C_T / 4)}$$

where C_T is the total strand concentration (typically 100 nM for PCR). For our GCATGC duplex, summing the tabulated ΔH and ΔS values (similar procedure to above) and assuming $C_T = 100$ nM gives $T_m \approx 23\,°C$. This means at room temperature, the duplex is just barely stable — consistent with our $K_d \approx 36$ μM estimate.

Why T_m matters. Every molecular biologist using PCR, hybridization, or oligo design uses T_m calculators built on SantaLucia parameters. The same parameters that predict T_m are the parameters that feed into the Boltzmann distribution for duplex formation — making the SantaLucia model the most experimentally validated application of the convergence equation in all of biology.

7. The Dot Product Structure

Now the key insight for the MRC framework. The SantaLucia energy is:

$$\Delta G = \Delta G_{\text{init}} + \sum_{i=1}^{L-1} \Delta G_{\text{NN}}(s_i s_{i+1} / s_i' s_{i+1}')$$

This is a sum over positions of a tabulated parameter. We can rewrite it as a **dot product** by introducing two vectors:

The parameter vector.

Define a 10-dimensional vector w_{NN} whose entries are the 10 tabulated NN parameters:

$$w_{\text{NN}} = (-1.00, -0.88, -0.58, -1.45, -1.44, -1.28, -1.30, -2.17, -2.24, -1.84)$$

(ordered as: AA/TT, AT/AT, TA/TA, CA/TG, GT/AC, CT/GA, GA/CT, CG/CG, GC/GC, GG/CC)

The sequence histogram vector.

For a given duplex, define a 10-dimensional vector x_{seq} where x_k = the count of NN pair k in the duplex.

For our GCATGC duplex:

- GC/CG (= GC/GC in table): count = 2 (positions 1-2 and 5-6)
- CA/TG: count = 2 (positions 2-3 and 4-5, the latter by symmetry)
- AT/AT: count = 1 (position 3-4)

- All others: 0

So $x_{\text{seq}} = (0,1,0,2,0,0,0,0,2,0)$.

The dot product.

$$\Delta G_{\text{NN}} = w_{\text{NN}} \cdot x_{\text{seq}}$$

Computing:

$$= (-1.00)(0) + (-0.88)(1) + (-0.58)(0) + (-1.45)(2) + (-1.44)(0) + (-1.28)(0) + (-1.30)(0) + (-2.17)(0) + (-2.24)(2) + (-1$$

$$= -0.88 + (-2.90) + (-4.48) = -8.26 \text{ kcal/mol}$$

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Same answer as Section 4 — now written as a dot product.

$$\Delta G_{\text{NN}} = w_{\text{NN}} \cdot x_{\text{seq}}$$

This is given bilinearity in its purest form. The binding energy is literally a dot product in raw sequence feature space. The parameters are physically interpretable (tabulated thermodynamic measurements) and the sequence features are physically interpretable (counts of dinucleotide pairs). No encoder, no learning — just physics.

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8. Connection to the Convergence Equation

Plugging the SantaLucia dot product into the Boltzmann distribution gives the **competitive binding probability** for a miRNA or TF seed region choosing among candidate target sites:

$$P(\text{bind target}_j) = \frac{\exp(-\Delta G_j / RT)}{\sum_k \exp(-\Delta G_k / RT)} = \frac{\exp(-(w_{\text{NN}} \cdot x_j) / RT)}{\sum_k \exp(-(w_{\text{NN}} \cdot x_k) / RT)}$$

Defining the score $s_j = -w_{\text{NN}} \cdot x_j / RT$:

$$P(\text{bind target}_j) = \frac{\exp(s_j)}{\sum_k \exp(s_k)} = \text{softmax}(s)_j$$

Comparison to transformer attention. The transformer attention for query q and keys $k_1, \ldots, k_M$:

$$A_j = \frac{\exp(q \cdot k_j / \tau)}{\sum_k \exp(q \cdot k_k / \tau)}$$

Under the identification:

- $q \leftrightarrow -w_{\text{NN}}$ (the agent's feature vector — here, the physics parameters)
- $k_j \leftrightarrow x_j$ (the target's feature vector — the sequence histogram)
- $\tau \leftrightarrow RT$ (the temperature parameter)

The SantaLucia model IS a special case of transformer attention. Written decades before transformers, with physical parameters in place of learned weights, it is the same mathematical operation.

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9. Higher-Order Refinements

The pure nearest-neighbor model is an approximation. Real DNA duplexes have known deviations:

Trinucleotide corrections. The SantaLucia 1998 parameters are already fit with dinucleotide resolution, but some third-nearest-neighbor effects remain uncaptured. These are small (~0.1–0.2 kcal/mol per position) and usually neglected.

Loop and bulge corrections. Mismatched base pairs, internal loops, and bulges contribute additional ΔG terms. SantaLucia & Hicks (2004, *Annu. Rev. Biophys. Biomol. Struct.*) tabulate these for the major loop/bulge types.

Salt dependence. ΔG depends on monovalent cation concentration, requiring corrections via $\Delta G([\text{Na}^+]) = \Delta G(1\,\text{M}) - 0.175 \cdot (L-1) \cdot \ln([\text{Na}^+])$ or similar empirical corrections.

Terminal modifications. 5'-phosphate groups, dangling ends, and non-Watson-Crick termini all have tabulated corrections.
The architectural mapping. Each correction has a direct mapping to a transformer architecture variant:

SantaLucia correction	Attention architecture variant
Nearest-neighbor (pure)	Vanilla attention
Trinucleotide	Local convolution on attention weights
Loop/bulge	Attention masking
Salt dependence	Learnable temperature τ
Terminal modifications	Special tokens (CLS, SEP)

These aren't coincidences — they are different physical corrections mapping to the same architectural solution because both are rooted in the convergence equation. See Methods for Molecular Recognition Computing (Reddy, 2026b) §S2.8 for a full derivation.

10. Summary

Concept	Result	Significance
Duplex energy	$\Delta G = \Delta G_{\text{init}} + \sum_i \Delta G_{\text{NN}}(i)$	Sum of per-position contributions
10 parameters	Tabulated from experiments	No learning required
Our worked duplex	$\Delta G = -6.30$ kcal/mol	6-bp GCATGC palindrome
Predicted K_d	~36 μM at 310 K	Marginally stable
Dot product form	$\Delta G_{\text{NN}} = w_{\text{NN}} \cdot x_{\text{seq}}$	Given bilinearity
Convergence equation	$P(j) = \text{softmax}(s)_j$ with $s_j = -\Delta G_j / RT$	Same as transformer attention
The identification	Tabulated params = "query"; sequence counts = "keys"	Physics = attention in raw space

11. Checkpoint Questions

Q1. Compute ΔG for the duplex 5'-GCGCGC-3' / 3'-CGCGCG-5' (an all-GC 6-bp duplex). Compare to the GCATGC example.
Answer: The 5 NN pairs are GC/CG, CG/CG, GC/CG, CG/CG, GC/CG. Looking up: GC/GC = −2.24 (×3), CG/CG = −2.17 (×2). Sum: $-2.24 \times 3 + -2.17 \times 2 = -6.72 + -4.34 = -11.06$ kcal/mol. Add initiation (+0.98 × 2 = +1.96): $\Delta G = -9.10$ kcal/mol. Compared to GCATGC at $\Delta G = -6.30$ kcal/mol, the all-GC duplex is 2.8 kcal/mol more stable (about 4.5 RT) — about 85× lower K_d. This is why GC-rich regions of genomes have higher PCR primer melting temperatures and why amplifying GC-rich templates requires specialized polymerases.
Q2. For a random 20-bp duplex with 50% GC content, estimate ΔG and K_d at 37°C.
Answer: 19 NN pairs. Average ΔG_{NN} across the 10 parameters is about -1.44 kcal/mol. For 50% GC, the distribution of NN pairs is roughly uniform among the 10 types (with slight enrichment of mixed pairs). So $\Delta G_{\text{NN}} \approx -1.44 \times 19 = -27.4$ kcal/mol. Adding initiation (+1.96 kcal/mol): $\Delta G \approx -25.4$ kcal/mol. At 310 K: $K_d = \exp(-25.4 / 0.616) = \exp(-41.2) \approx 10^{-18}$ M (attomolar). A 20-bp duplex is extraordinarily stable — this is why longer PCR primers anneal reliably and why single-molecule FISH probes work.
Q3. A student argues: "The SantaLucia parameters are empirical fits to experimental data. They're not derived from first principles. So calling this 'given bilinearity' is an overstatement — it's really just a curve fit."
Answer: The student identifies a real subtlety but draws the wrong conclusion. The SantaLucia parameters ARE empirical, but the DECOMPOSITION of the energy as a sum over nearest-neighbor

pairs is a **structural claim** that follows from the physical nature of DNA stacking — each base pair's thermodynamic contribution depends dominantly on its immediate neighbors, not on the global sequence. This structural claim is a physical hypothesis that has been tested thousands of times across millions of duplexes and holds to within 0.5 kcal/mol accuracy. The numerical values of the 10 parameters are empirical; the bilinear form is physical. For the MRC framework, what matters is the bilinear form (that's what makes it a dot product), not the precise parameter values. The fact that the physics prescribes this specific form is exactly what "given bilinearity" means.
Q4. In what sense is the SantaLucia model different from position weight matrices for TF binding?
Answer: Both are given bilinearity — both decompose binding energy as a sum over positions, making each a dot product. The differences: - **Feature space:** SantaLucia uses nearest-neighbor dinucleotide features (10-dimensional vector of pair counts). PWMs use single-nucleotide positional features (4L-dimensional vector of one-hot encodings). - **Parameter source:** SantaLucia parameters come from thermodynamic measurements of DNA duplex stability. PWM parameters come from binding-site enrichment experiments (SELEX, ChIP-seq). - **Scope:** SantaLucia applies universally to all DNA duplexes with the same 10 parameters. PWMs are TF-specific — a different PWM for each TF.
Both fit the convergence equation. Both predate transformers by decades. Next whiteboard (WB2) will work through the PWM derivation explicitly.

12. Connection to Future Lectures

- **Lecture 5 WB2:** Position weight matrices are the same type of dot-product decomposition applied to transcription factor binding. The central derivation of Lecture 5.
- **Lecture 5 WB3:** Emergent bilinearity — when the dot product exists only in a learned embedding space (antibody-antigen).
- **Lecture 6:** Luce's theorem proves that the softmax form applied to these dot products is the unique selection function satisfying five axioms.
- **Lecture 7:** The full SFM architecture is derived. For given-bilinearity domains (miR-SFM, tSFM, crisprSFM), the encoder implements the SantaLucia-like decomposition. For emergent domains (CALM, TCR, enzyme, drug-target), the encoder must learn it.
- **Lectures 9–11:** Students building miR-SFM will use RNA-FM or DNABERT-2 as encoders, exploiting given bilinearity in the miRNA-mRNA system. Faster convergence is expected, consistent with the vibe-coding paper's result: miR-SFM achieves R@1 = 98.4%, the highest among the six prototypes.